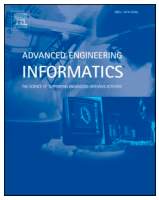
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Full length article



Intervention and management of construction workers' unsafe behavior: A simulation digital twin model

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ABSTRACT

Efforts to address unsafe behaviors of construction workers have predominantly focused on identifying causative factors, unraveling formation mechanisms, and devising pre-emptive control strategies. Nonetheless, these studies have inadequately examined the reasons behind workers' engagement in unsafe behavior from a behavioral psychology perspective, specifically the collision reaction between impulsivity and self-control in the decision-making process, as well as the lack of exploration of psychological processes and behavioral response mechanisms of workers under different management strategies on construction sites. This paper describes an explanatory closed-loop structured model to simulate the safety performance of workers' behavior under different intervention strategies, considering the behavioral decision process and interactions between workers and managerial environment. The hybrid simulation model for worker's unsafe behavior consists of three parts: the control panel, the proactive management panel, and the reactive panel. The model provides valuable information for explaining the causation of workers' unsafe behaviors and finding efficient management strategies. Through simulation experiments, results show that proactive management has more advantages in maintaining workers' behavioral safety levels over a long period of time than reactive management. This research provides valuable information for policy analysis and decision-making for project managers.

1. Introduction

The concept of a 'Digital Twin' encapsulates a symbiotic relationship between the tangible physical universe and its abstract digital counterpart [1]. Within the sphere of construction activities, the very essence of a work site is distilled into core components, namely: the workforce, material resources, infrastructural facilities, equipment, and the encompassing landscape [2]. These constituents play a pivotal role not only in the physical world but also significantly contribute to creating digital models. The digital model is established by abstracting the characteristics and behaviors of these critical elements from the physical world and using computer technology for simulation [3]. This process typically includes the collection of actual data relating to these components, worker productivity, equipment performance parameters, material physical properties, facility layout, and geographical features of the landscape, and then converting this data into digital representation through data analysis techniques [4]. This digital model serves as a powerful tool to elucidate and delve into the root causes behind real-world phenomena, thereby providing valuable insights to aid strategic decision-making for managing real-world scenarios [5].

Extensive research has already been undertaken within the construction realm, concentrating on digital twins of various elements such as Building Information Modeling (BIM) for site facilities [6], Geographic Information Systems (GIS) for landscapes [7], and System Information Modeling (SIM) for electrical equipment [8]. However, it is essential to recognize that workers, who form the backbone of construction activities and significantly influence the efficiency and safety of project execution, have been comparatively less studied in terms of their digital twin representation, particularly in the interaction mechanisms between management strategies and worker behavior. Given their actions can potentially lead to accidents [9], focusing research efforts on their digital representation is of paramount importance. According to Heinrich's study, it was discovered that unsafe practices were the main culprits behind 88% of industrial accidents in a sample of nearly 75,000 accidents [10]. Further supporting this notion, A study by Salminen and Tallberg [11] found that 84–94% of accidents resulting in fatal

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occupational injuries and major casualties were caused by human errors. These insights underscore the heightened importance of addressing worker safety in the construction sector. According to a report from the U.S. Bureau of Labor Statistics in 2019, construction-related fatalities accounted for over a fifth of the total 5,333 recorded workplace fatalities [12]. Moreover, the prevalence of non-fatal injuries and illnesses attributable to falls, slips, and trips stood at 31.4 per 10,000 full-time workers in the construction sector, dwarfing the average figure of 21.7 across all private sectors [13]. These data vividly illustrate the urgent need to prioritize digital modeling research on workers' unsafe behaviors to gain a deeper understanding of the underlying causes and seek effective solutions through modeling. As shown in Fig. 1, an Agent-Based model (ABM) of workers' unsafe behaviors is proposed to simulate and explain their behavioral responses and decision-making under varying work environment conditions in this paper. This modeling approach provides crucial support for future decision-making, such as the implementation of more stringent safety regulations or the enhancement of worker training programs.

Many studies have focused on the causation of workers' unsafe behaviors and the corresponding effective countermeasures. However, most of the existing studies neglected the complex adaptive nature of construction workers' unsafe behaviors. In other words, previous studies analyzed the causations of workers' unsafe behaviors from a static and macro perspective. For example, Fang (2008) [14] and Dodoo (2019) [15] et al. investigated the factors driving construction workers to engage in unsafe behavior. Wang et al. (2022) [16] demonstrates a clear connection between construction accident prevention and unsafe behaviors.

Although some scholars have begun to explore the causal mechanism of workers' unsafe behaviors based on the system dynamic methods [17], few studies have investigated the dynamic and micro mechanisms at the root of workers' personal decisions to engage in unsafe behavior. In other words, the causation of workers' unsafe behaviors involves management effects, a psychological process and a decision-making mechanism. The existing research lacks a complete perspective to sort out the relationship between the above three aspects.

Actually, the management of workers' unsafe behaviors and their subsequent reactions to the management measures can be regarded as a dynamic and feedback loop process. First, human behaviors can be viewed as a response to external stimuli [18], and it is not static or without provocation. External factors, including the stimulation of the poor working environment and management measures, are usually the direct causes of unsafe behaviors of workers. Second, given external stimulation, human behavior has a complex psychological process, i.e., weighing the pros and cons and consequences of different behavior choices [19]. The final behavior choice is determined under the psychological fighting of different motives. Third, by observing the behavior of workers on the production site, site managers can analyze the management performance of different management policies. Furthermore, management staff continually adapt their management strategies to maintain the site's safety performance within a reasonable range [20].

Based on the above assumptions, this study presents a closed-loop structured model to understand and explain workers' unsafe behaviors, as shown in the digital model in Fig. 1. In this model, emphasis will be placed on a dynamic feedback process: Site management staff observe the behavioral safety performance of the entire site and carry out statistical analyses. If the management personnel find that the behavior safety performance of the current construction site is too poor, they will put forward strict management disciplines to avoid safety accidents. As part of rigorous management measures, unsafe worker behaviors will be reduced due to high pressures, judge and weigh. As a result, the safety performance of the construction site will be improved and maintained for a period of time. To gain more management knowledge about this process, hybrid simulation is an effective approach to obtaining insights.

Workers' unsafe behavior is an interdisciplinary field that encompasses various aspects of civil engineering construction, including specific management measures and the working environment [21]. However, its foundation can be traced back to the realm of behavioral psychology [22]. Previous research has inadequately addressed the reasons behind workers' engagement in unsafe behavior from a behavioral psychology perspective, particularly the collision reaction between impulsivity and self-control in the psychological decision-making process that leads to such behavior [23]. In addition, there is a lack of exploration on the psychological processes and behavioral response mechanisms of workers under different management strategies on construction sites. To fill this research gap, our study integrates various behavioral psychology theories and employs an agent-based composite simulation model, which enhances the interpretability of results and facilitates the exploration of multiple scenarios. By integrating simulation with real-world data, we provide valuable insights into the factors influencing unsafe behaviors among construction workers. The findings underscore the significance of proactive safety management and offer practical implications for improving safety performance and developing effective safety management strategies.

This paper builds upon existing ABM research on worker behavior and further incorporates classic theories of behavioral psychology into the worker psychological decision-making process, enhancing the theoretical foundation of relevant research. Additionally, the computational experimental results of this paper explain the phenomenon of differences in behavior safety performance under different management strategies, providing project managers with a highly interpretable scenario simulation tool that is conducive to the development of safety management strategies.

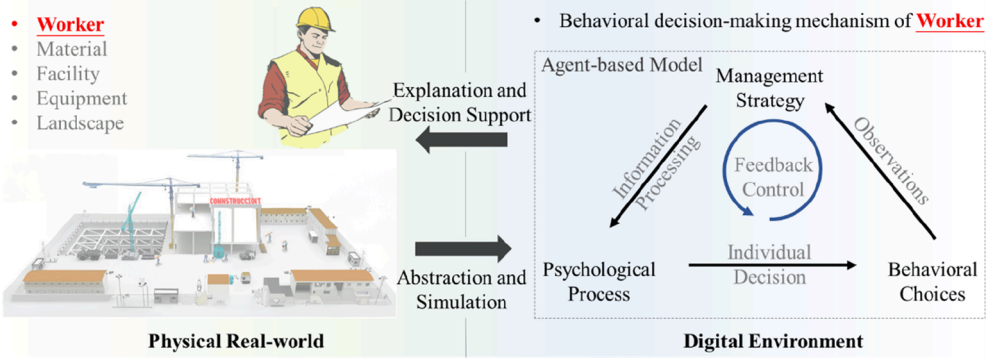


Fig. 1. Digital-twin model of construction workers.

2. Related research in construction unsafe behavior management

2.1. Behavior-Based safety (BBS) theory

Behavior-Based Safety (BBS) [24] emphasizes the procedure from behavior observation to safety management. The BBS is a process of educating management and employees on the overall safety of the work place through behavior observation [25]. To improve workers' behavioral safety performance, BBS is a common method that attracts great attention. This is a management process, where supervisors were responsible for monitoring worker behavior, providing feedback, and implementing positive or negative interventions.

The procedure of BBS consists of four parts [25], i.e., objective identification (what should workers not do?), behavior observation (what are workers doing?), causation analysis (why workers do it?), and safety communication (intervention strategy to improve what workers do) [26]. During the implementation of the BBS, a checklist of unsafe behaviors to be observed in the field should be established first. Employees who have received on-site safety review training are selected as observers. They will review and supervise other employees, including paying attention to their safety or unsafe behaviors and making records. Based on the observation records, the Antecedent-Behavior-Consequence (ABC) model is generally used for causation analysis [27]. This process involves tracing and inferring observed behaviors, attempting to re-establish the causes of unsafe behaviors, and speculating on their potential outcomes, as a basis for safety education. Given the observational and analytical data, safety communication and intervention strategies will be implemented, including the positive and negative reinforcement stimulus.

The BBS management, as an important pre-control means of safety management, was largely concerned and discussed in scientific research and engineering practice. One of the most important issues facing BBS management is that once intervention actions are removed, the effect of improved behaviors is difficult to sustain [28].

2.2. Influencing factors

The BBS pays greater attention to behavioral observation, and many other studies explore the relationship between safety management and unsafe behaviors.

Clarke et al. (2000) [29] clarifies the term safety culture (or named safety climate), and points out that organizational safety culture has an indirect but far-reaching impact on workers' unsafe behaviors. The development of safety culture on the construction site includes three levels from shallow to deep, including the norms and artefacts (surface level), the beliefs and values (intermediate level), and the core assumptions (deepest level, i.e., understanding that safety is the overriding priority). Choudhry et al. (2006) [30] discussed how safety culture improves workers' behaviors through increasing their safety knowledge, motivation, compliance, and participation. A good safety culture positively impacts safety managers (senior managers, line managers, and supervisors) and frontline production staff.

In discussing the relationships between the unsafe behavior and safety management, some researchers have developed a causation model to explain the impacts of safety management on workers' unsafe behaviors. Jiang et al. (2015) [17] presents cause-effect relationships between management conditions on safety and workers' individual conditions, in which workers' safety awareness, safety knowledge, safety attitude, subjective norms, and perceived behavioral control are considered. Guo et al. (2018) [31]presents a behavior-based safety program to simulate the impacts of different management measures, i.e., rewards and punishments, on workers' unsafe behaviors. Yu el al. (2019) [32] proposed a system dynamic model for intervening workers' unsafe behaviors, in which the impact of individual factors, the physical environment, safety leadership, group factors and safety management on the development of unsafe worker behaviors are considered.

The above research considered the inhibition or promotion of various management measures and related factors on unsafe behaviors. However, little theoretical or methodological support in behavioral psychology is taken into account in these researches, and more details of mental activities from an individual level is needed.

2.3. Formation mechanism

Regarding to people's psychological process and their consequential behaviors, it has a long history in the theories and research, which is called behavioral psychology [33]. Some studies have been conducted to model people's behaviors using simulation-based methods, in which the conceptual decision-making process is developed [32,34].

In the domain of behavioral psychology, behavior can be considered as an individual activity adapting to the environment, and its formation has reasons and motivations. Kurt Lewin (1890–1947), usually called the father of social psychology, established the Lewinian Field Theory, which stressed that behavior is a function of the person and the environment [35]. External environment factors, including management policies, working environment, physiological feelings, etc., are the direct drivers of people's behaviors. But environmental stimuli are not the only factors shaping behavior. Even under the same environmental stimulus, different people will have their own behavioral choices as a result of their personalities.

On the basis of Lewinian Field Theory, Wilson et al. (2000) [36] proposed a dual attitude model to explain the formation of people's social behaviors. The dual attitude theory assumes that people can have two completely different attitudes at the time of making a certain behavioral decision: one is the automatic and impulsive attitude, and the other is the conscious and reflective attitude.

2.4. Agent-based modeling

The main components of an ABM include agents, the environment in which agents operate, and the interactions between agents and their environment as well as with each other [37]. It is also necessary to define rules that govern the communication, behavior, and decision-making capabilities of agents. The heterogeneous nature of ABM allows for differentiation among individual agents and the generation of various scenarios [38]. ABM are an efficient solution for dealing with multi-option problems [39].

The advantages of ABM have led to their application in construction safety management. Some studies focus on a macroscopic level, where safety interactions between companies, employees, the public, and the government in small enterprises are explored [40]. Agents, such as contractors, projects, insurance companies, governments, and the environment, are considered to simulate interactions between stakeholders at the market level with regards to safety [41]. In order to study safety performance, interactions between a worksite, individual construction workers, and different safety investments were mimicked [42]. The safety behavior of various contractors involved in a project and its impact on project performance were modeled [43].

ABM have also been used to study interactions among organizational factors, supervisors, and workers at the project level. Newaz et al. (2020) [44] examined how supervisors' behavior impacts the safety behavior of workers using the concept of psychological contract. The social contagion effect of safety violations within construction crews was explored, considering cognitive, social, and organizational aspects [45]. A cognitive analysis model was developed to simulate worker behaviors and on-site interactions between construction workers and management teams [46]. ABM were employed to investigate how the socio-cognitive process and safety management interventions interact and influence workers' safety behaviors under different site risk conditions [47]. Raoufi et al. (2018) [48] examined the influence of crew motivational factors, crew situational factors, and project situational factors on

performance. The development of an ABM to explore the impact of changes in workgroup dynamics on the socio-cognitive process of workers' safety behavior was presented [49]. A complex system of unsafe behavior, including cognitive processes, individual cognitive factors, social organizational factors, and social groups, was developed [50].

Summary:

Existing research primarily focuses on analyzing the factors that influence worker unsafe behaviors (causes), understanding the mechanisms underlying the formation of unsafe behaviors (formation process), and implementing pre-control methods for unsafe behaviors using intelligent tools (such as intelligent identification tools and simulation tools based on ABM). However, there are still opportunities for improvement in understanding workers' behavioral decision-making mechanisms and analyzing management strategies.

On one hand, current research heavily relies on risk perception theory to model the formation process of worker unsafe behaviors. This theory assumes that workers are rational decision-makers who always choose the most reasonable course of action based on risk assessment. However, in reality, people's behavior can be influenced by irrational factors and impulsive actions due to factors like poor emotions, high schedule pressure, and physical conditions [51].

On the other hand, the behavioral safety performance of construction site workers is the result of the combination of top-level management strategy implementation and bottom-level worker behavior decision-making and response. Different management strategies can trigger different psychological processes and coping response styles in workers. For example, some workers may engage in safe behavior voluntarily due to a high level of safety attitude driven by a well-established proactive control mechanism, while others may engage passively in safe behavior due to strict reward and punishment measures [52].

The background presented above introduces two main objectives of this research. Firstly, to deepen the understanding of the formation mechanisms behind worker unsafe behaviors by integrating the theory of dual attitudes derived from behavioral psychology. This integration will provide interdisciplinary theoretical support rooted in psychology for research in related fields. Secondly, to address the sustainability issues of on-site unsafe behavior management effectiveness, specifically focusing on the observed rebound in the incidence of unsafe behaviors after stopping intervention [53].

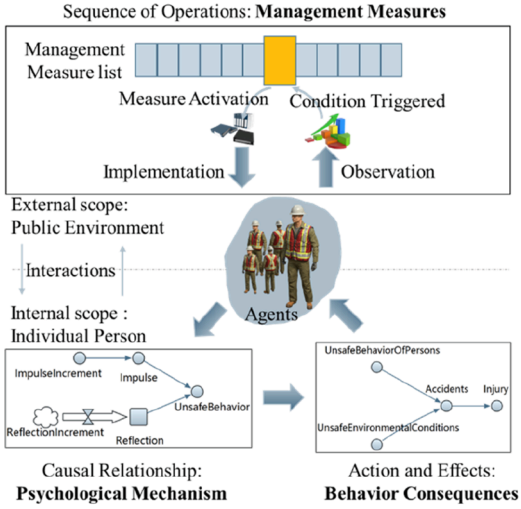
3. Model development

3.1. Modeling methods: Hybrid simulation modeling

The general framework for simulating construction workers' unsafe behaviors is depicted in Fig. 2(a). This framework comprises the complex internal activities (internal scope) of individuals and the interactions between individuals and the environment (external scope). To accomplish this, the adopted approach in this study utilizes the hybrid simulation modeling technique, which relies on the ABM framework integrating both system dynamics and discrete event simulation. The reasons for selecting these methods are as follows: they are widely utilized modeling and simulation techniques for studying and analyzing the dynamic processes of complex systems, and they offer quantitative analysis of system behavior and performance. Nevertheless, each method possesses its distinctive characteristics and applicability in specific scenarios.

ABM [54] is a modeling technique well-suited for individual representation. Every agent is endowed with unique attributes, state, and behavior rules, with their state and behavior being influenced by the environment [55]. In the present study, by simulating the behavior of a considerable number of agents, ABM enables the examination of the emergence and evolution of overall system behavior, as well as comprehension of the ramifications of individual decisions on system performance. System dynamics [56] concentrates on the evolutionary behavior of the entire system and captures changes in system state using continuous-time dynamic equations. It underscores the interactions and feedback mechanisms among different components of the system, rendering it suitable for modeling "*causal relationships*". In this study, individual behavior is regarded as the outcome, whereas the individual's physiological and psychological environment, along with safety attributes, are considered as influencing factors. Modeling this intricate causal pathway and influence mechanism from influencing factors to the result aligns with the applicability and suitability of the system dynamics method. Discrete event simulation [57] deals with systems where the state changes occur at specific time points triggered by "*events*". In our study, the management measures implemented on the construction site adapt automatically based on the behavioral safety performance. For instance, if a safety accident occurs, stricter measures are enforced. Therefore, modeling the impact effects of these management measures aligns with the applicability of discrete event simulation.

In summary, ABM is useful for developing the framework to simulate the interactions between individuals and the overall environment,



(a) General Framework (Section 3.1)

Research Content		Modeling Method	Software
Proactive management modeling (Section 3.2.1)	Dual-attitude decision system	System dynamic	Anylogic
	Accident causation process	System dynamic	
	Proactive management measures	Agent-based Modeling + Discrete event modeling	
Reactive management modeling (Section 3.2.2)	Expected reward-based behavior incentivization	System dynamic	
	Accident causation process	System dynamic	
	Reactive management measures	Agent-based Modeling + Discrete event modeling	

(b) Two different management strategies (Section 3.2)

Fig. 2. The overall idea to simulate construction workers' unsafe behaviors.

system dynamics is useful for modeling "causal relationships" (i.e., individual internal modeling), and discrete event simulation is useful for modeling "condition-triggered events"(i.e., management measures modeling).

With the overall framework in place, the safety performance of various strategies of behavioral safety management can be simulated and compared. These strategies encompass proactive management and reactive management, as illustrated in Fig. 2(b). The AnyLogic software tool is employed to model workers' unsafe behaviors in both management strategies. In proactive management, workers primarily experience internal conflicts between impulsive and reflective cognitions and make behavior decisions through a dual-attitude decision model. In contrast, in reactive management, workers primarily consider the punishment and loss resulting from their unsafe behavior and make behavior decisions based on expected rewards.

3.2. Relevant models for unsafe behaviors management

3.2.1. Proactive management modeling

Framework of the discrete choice model.

The Proactive Safety Management Strategy ensures the psychological construction of reflective cognition of workers in terms of behavioral safety on a regular basis. Multiple intervention measures can be implemented to awaken and improve the reflective attitude of workers towards behavioral safety, including safety training, safety education, etc. Under the proactive strategy of safety management, workers' behavioral decision is mainly developed based on the dual attitude process.

As shown in Fig. 3, according to the dual attitude theory, the unsafe behavior of construction workers is actually a choice behaviour [58], consisting of a mental process of judging the merits of multiple options and deciding whether to choose one for action. More specifically, the choice that workers face is that of complying with the impulse to make an unsafe behavior or obeying reflective thinking to make a safe behaviour [59]. In the context of choice behavior, workers' decision of unsafe behavior is simulated using the discrete choice model [60]. Discrete choice model is developed based on the random utility theory, which assumes that when presented with a discrete set of options, people will choose the option with the greatest benefit or utility.

By combining the discrete choice model with the dual attitude theory, the probability ratio of a worker conforming unsafe behaviors to safe behaviors can be predicted using the logit model, which is detailed as Eq. (1):

$$\frac{p_u}{1-p_u} = e^{U_r - U_i} \tag{1}$$

where p_u denotes the probability of unsafe behaviors for a worker, U_r denotes the utility of his/her reflective cognitions, and U_i denotes the utility of his/her impulsive cognitions. The calculations of U_r and U_i are based on Eq. (3) and Eq. (4), detailed example can be referred to Table 1. Considering that $U \in [0, 1]$, a truncated logit model is used to correct the inaccuracy in Eq. (1). Thus, the maximum value of p_u should be $p_{u,max} = 1/(1+e^{-1})$, and its minimum value should be $p_{u,min} = 1/(1+e^{1})$.

$$p_u = \frac{1/(1+e^{U_r - U_i}) - p_{u,min}}{p_{u,max} - p_{u,min}} \times \mu_{U_r} \tag{2}$$

where $\mu_{U_r} = (1-U_r)^{\tau}$ is a dominant coefficient of reflective cognitions, and $\tau \in [0, 1]$. If $\tau = 1$, it indicates that the reflective cognitions have a thorough dominating effects over the impulsive cognitions. If $\tau = 0$, it indicates that the reflective cognitions have no dominating effects over the impulsive cognitions. Considering reflective cognitions can override impulsive cognitions when it comes to behavior decision [61] if the worker has strong self-control capacity, we can take $\tau = 1$ in such a situation.

Modeling of the Reflective attitude and Impulsive attitude.

The process of forming a worker's reflective attitude and impulsive attitude is the key to modeling unsafe behaviors. According to the dual attitude theory [62], the hot system (i.e., impulsive attitude) is activated by the external environment automatically without much thinking, and the cool system requires a complex process in consideration to invoke the self-control procedure.

Thus, the formation of impulsive attitude U_r is straight-forward and does not use working memory, so the linear functions are used here to model the causal relationship between his impulsive cognitions and the influential factors. The detailed example can be referred to Table 1.

In contrast, the formation of reflective attitude is much more complex since the memory system of the worker is used and has to be modelled. Specifically, the memory system of a worker in terms of behavioral safety decision, includes the learning process of memory and the forgetting process of memory. In learning curve theory [63], there is a sigmoid relationship between the effort devoted to learning and the resultant performance improvement. Thus, the learning function can be written as

$$y = \frac{1}{1+e^{-(x-a)}} \tag{3}$$

where y denotes the resultant strength of memory (e.g., the safety awareness, or safety proactivity), which is an important component of the reflective cognitions U_i. The detailed example can be referred to Table 1. $\times$ denotes the effort devoted to learning (e.g., the number of safety management measures implemented), a is a predefined parameter describing the worker's characteristic. The theory of the forgetting curve [64] is that information learned will disappear from our memories over time unless we take measures to preserve it. Thus, the memory of the reflective attitude needs to be gradually built up with the process of accumulating and strengthening, and will be gradually forgotten with the progress of time. Suppose the strength of memory following the learning process of Eq. (3) is y_{max}, the memory retention rate after a time period t can be calculated by using the Ebbinghaus forgetting curve [65]:

$$y(t) = y_{max} e^{-\frac{t}{s}} \tag{4}$$

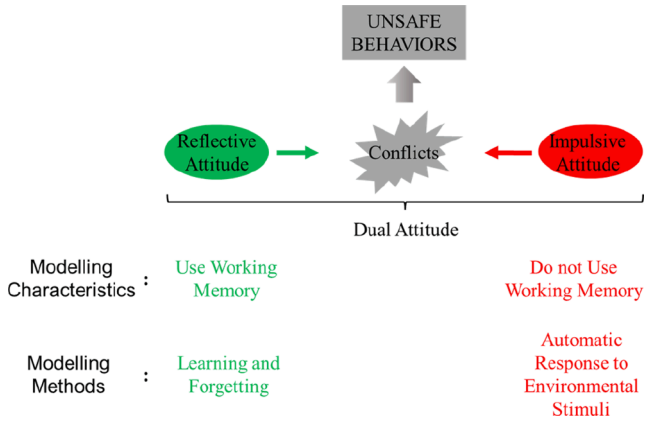


Fig. 3. Dual attitude conflicts and modeling methods.

3.2.2. Reactive management modeling

The Reactive Safety Management Strategy mainly uses reinforcing and incentive policies to ensure the behavioral safety on construction site. Positive reinforcement (e.g., award) and negative reinforcement (e.g., penalty and possibility of discharge) are common measures used to guide and shape an encouraged behavioral style during interventions. In such circumstances, the behavioral decision of a worker agent follows a Reinforcement Learning (RL) [66] paradigm. The worker agent is assumed to make his decision with the goal to maximize his expected reward in time step t, i.e.,

$$E(R_{t+1}|A_t) = -P_t \bullet A_t - U_{i,t} \bullet (1 - A_t) \tag{5}$$

Table 1
Equations used in the proactive management panel.

Module	Model variables	Equations
Reflective Precursor module(Theory basic: [63,64])	$y_{g,t}$ (Growing rate of y_t)	Suppose $f(x) = \{1 + exp[-(x-a)]\}^{-1}$, and its inverse function is $f^{-1}(y_t) = g(y_t) = a - ln(1/y_t - 1)$. Then, $y_{g,t} = f(g(y_t) + \Delta x) - y_t$where $\Delta x = \sum_{i=1}^{6} \alpha_i \bullet x_i$, x_i denotes the influence factors, such as Safety Meeting, Safety Training, etc. α_i is the linear coefficient parameter of x_i.
	$y_{d,t}$(Decreasing rate of y_t)	$y_{d,t} = y_{max}\left(1 - \frac{1}{s}e^{-\frac{t}{s}}\right)$
	y_t(Variable value of Safety Awareness, Safety Proactivity, or Emotional Exhaustions at time t)	$y_{t+1} = y_t + y_{g,t} - y_{d,t}$
	ReflectiveCognitions	*ReflectiveCognitions* = *SafetyAwarenessÂ·SafetyProactivityÂ·*(1 − *EmotionalExhaustions*)
Impulsive Precursor module (Theory basic: [62])	q_t(Variable value of Unfavorable Physical Condition, or Unfavorable Mood at time t)	$q_t = \sum_{i=1}^{6} \beta_i \bullet v_i$, $v_i \in \{0,1\}$ is the binary variable indicating the perturbation factors, such as Headache, Heat Exhaustion etc. β_i is the normalized linear coefficient parameter of v_i, which satisfies that $\sum_{i=1}^{6} \beta_i = 1$.
	ImpulsiveCognitions	*ImpulsiveCognitions* = *UnfavorablePhysicalConditionÂ·UnfavorableMood*
Accident Generation module (Theory basic: [60,69])	*UnsafeBehaviorProbability*	$UnsafeBehaviorProbability = = \frac{1/(1+e^{ReflectiveCognitions - ImpulsiveCognitions}) - p_{u,min}}{p_{u,max} - p_{u,min}} \times (1 - ReflectiveCognitions)$where $p_{u,max} = 1/(1+e^{-1})$, $p_{u,min} = 1/(1+e^{1})$.
	UnsafeBehaviorIndicator	$UnsafeBehaviorIndicator = \begin{cases} 1, rand() \le UnsafeBehaviorProbability \\ 0, rand() > UnsafeBehaviorProbability \end{cases}$
	Environmentalhazardintheworkplace	$Environmentalhazardintheworkplace = \begin{cases} 1, rand() \le Environmenthazardlevel \\ 0, rand() > Environmenthazardlevel \end{cases}$
	ExposuretoEnvironmentalhazard	$ExposuretoEnvironmentalhazard = \begin{cases} 1, rand() \le Exposureprobability \\ 0, rand() > Exposureprobability \end{cases}$
	NearMissAccidentIndicator	*NearMissAccidentIndicator* = *Environmentalhazardintheworkplace* • *UnsafeBehaviorIndicator* • (1 − *ExposuretoEnvironmentalhazard*)
	MinorinjuryIndicator	If *rand*() ≤ *Minorinjuryratio*, then *MinorinjuryIndicator* = *Environmentalhazardintheworkplace* • *UnsafeBehaviorIndicator* • *ExposuretoEnvironmentalhazard*; Else *MinorinjuryIndicator* = 0.
	MajorinjuryIndicator	If *rand*() > *Minorinjuryratio*, then *MajorinjuryIndicator* = *Environmentalhazardintheworkplace* • *UnsafeBehaviorIndicator* • *ExposuretoEnvironmentalhazard*; Else *MajorinjuryIndicator* = 0.

where $E(R_{t+1}|A_t)$ is the expected reward for a worker resulting from his action A_t. A_t is an unsafe-action binary indicator ($A_t = 1$ if he takes an unsafe behavior in time step t, and $A_t = 0$ otherwise). P_t is the cumulated discounted penalty for the workers' unsafe behavior, which can be calculated as

$$P_t = p_t + \gamma \bullet P_{t-1} \tag{6}$$

where $\gamma \in [0,1]$ is the discount factor in the RL algorithm, and p_t is the instantaneous penalty for the workers' unsafe behavior, i.e.,

$$p_t = (1-s)B_t + s \bullet A_t \tag{7}$$

where B_t is the major incident indicator ($B_t = 1$ if the worker is severely injured in time step t, and $B_t = 0$ otherwise). The process of workers' unsafe behaviors leading to safety accidents can be illustrated by referring to Fig. 5. $s \in [0,1]$ is the strictness level of safety management. Obviously, s reflects the project leaders' awareness of the importance of behavioral safety in construction operations. If $s = 0$, there will be no sanctions against workers unless a major incident occurs ($p_t = B_t$). If $s = 1$, the worker will be punished once he is found to have unsafe acts ($p_t = A_t$).

In Eq. (5), $U_{i,t}$ refers to the impulsive cognition level of the worker at time t, which has been used in Eq. (3). Eq. (5) is developed based on such an assumption that a worker will suffer different types of loss corresponding to his behavioral choice (safe behavior/unsafe behavior). If the worker behaves in a safety manner, he has to inhibit his psychological impulse and lose the opportunity to obey his inner desire to do what he wants to do. Thus, the impulse amount should be the loss of a worker with safe behavior. Otherwise, if the worker engages in dangerous behavior, the loss should be the amount of punishment (i.e., cumulated discounted penalty) imposed by the project managers.

According to the RL algorithm, the worker agent will take an action that maximize his rewards (or minimize his losses), which can be formulated as

$$A_t = argmax(E(R_{t+1}|A_t)) \tag{8}$$

As shown in Fig. 4 (b), a feedback procedure is designed to control the safety performance of on-site workers. No intervention will be implemented unless the undesired event (e.g., the unsafe behaviors or the incidents) has happened. In contrast, in the proactive management mode (as shown in Fig. 4(a)), the safety intervention will be implemented to increase worker awareness of behavioral safety in prevention, even if there are still no adverse events.

Given the unsafe behaviors of workers, its following consequences should be analyzed using the Heinrich's safety triangle theory [67]. The safety triangle model assumes a stable ratio of unsafe behaviors, near misses, minor injuries, and major injuries in the workplace. In other words, depending on the safety conditions on the site, the unsafe behavior of the workers has some probability and will still evolve into a major safety accident. To elucidate this causation process, the most referenced model, Heinrich's Domino Theory [68], is employed in our study. This theory outlines the progression from unsafe worker behavior to the occurrence of safety accidents, much like dominoes falling in a sequence. It identifies key factors including unsafe acts, hazardous working conditions, accidents, and resultant injuries. Integrating both Heinrich's Safety Triangle Theory and Domino Theory as our foundational tenets, as depicted in Fig. 5, this paper posits that safety accidents result from a blend of risky worker actions and perilous environmental conditions. If a worker is in spatial–temporal proximity to an accident, they may get injured, with a higher likelihood of minor injuries and a lower probability of major ones. These probability parameters can be estimated using expert knowledge.

4. Model validation through empirical data analysis

4.1. Procedures of model validation

The purpose of this study is to build a simulation model that

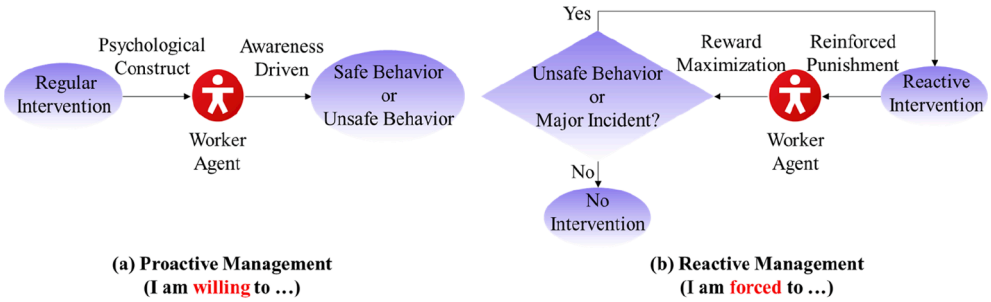


Fig. 4. Procedures of different behavioral safety management mode.

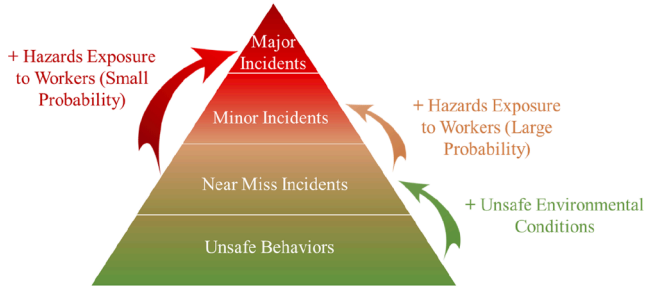


Fig. 5. Accident causation process resulting from workers' unsafe behaviors.

accurately simulates the psychological decision-making process of workers with unsafe behaviors. Specifically, the model seeks to replicate the inner game and decision-making processes of workers, including the key motivation types that influence their behavior and the ways in which these motivations interact. Considering that relying solely on observation to reconstruct unsafe behavior patterns and psychological activities can be challenging. Workers may alter their behavior when being monitored, making it difficult to capture their true safety practices.

To address this issue, a Survey Questionnaire was employed to investigate the impact of external factors and workers' internal psychological activities on their safety behaviors during construction. The questionnaire design has two key advantages: 1) *Anonymity* - anonymous survey questionnaires were utilized to investigate the psychological activities underlying unsafe behaviors among workers in different job types. This approach reduces psychological stress on respondents and provides more reliable data reflecting their actual psychological activities. 2) *Scenario-based Immersion* - to achieve the research objective, the questionnaire includes various hypothetical work scenarios that prompt respondents to immerse themselves in real-world situations and carefully review their decision-making process.is that it fully takes into account the subjective feelings and experiences of the respondents, thereby improving the reliability and validity of the data.

We distributed questionnaires to 90 construction workers with different job types and working on different projects, and conducted data analysis after the questionnaires were collected. This method allows for the collection of a large amount of data in a short period of time and covers participants with different working backgrounds and experiences, enhancing the representativeness and generality of the study. As shown in Appendix, the designed questionnaire consisted of three parts (30 questions in total), including external environmental factors, management strategy influences, and overall models. Respondents used a scale of 0 to 10 to indicate their trend for each question, 0 indicating 'not at al', 5 indicating 'somewhat possible, but not certain', and 10 indicating 'definitely'. Open-ended questions were also included in the questionnaire, i.e., Questions 29–30.

4.2. Results

We collected 90 survey questionnaires and received 84 valid responses from workers. The responses were analyzed and visualized using statistical methods to validate the proposed model's reliability.

The collected questionnaire data underwent statistical analysis to assess its reliability and validity. *The Cronbach alpha coefficient* has been calculated, which measures the internal consistency between the survey questions. The obtained value of 0.868 exceeded the threshold of 0.7, indicating a high degree of internal consistency and that respondents provided consistent answers to the questions related to the same topic. This suggests that collected data from survey questionnaire is reliable, and the results from the survey are valid. As illustrated in the right portion of Fig. 6, the reliability of our data is further bolstered by the diverse range of respondents spanning various occupations, educational levels, and project types.

The test–retest reliability correlation coefficient of 0.0225 suggests low stability between two respondents, with significant individual differences in workers' decision-making regarding behavioral safety, as evidenced by Figs. 6 and 7. The average coefficient of variation of 34.4% and an average standard deviation of 2.17 across all question ratings imply a high level of variability in the responses of workers.

From the above two statistical indicators, it can be observed that although workers have coherent decision-making patterns and mechanisms when it comes to unsafe behavior, the differences in decision-making mechanisms among individual workers are very significant. This reflects the correlation and personalization of workers' unsafe behavior.

- According to the responses of workers to Questions 1–19, we found that: *I)* Workers' scores for Questions 1–9 varied significantly, with approximately 74% acknowledging that physical and psychological disturbances in the work environment have a considerable impact on their decision-making process regarding behavioral safety, as depicted in Fig. 7. In contrast, around 26% of workers believed that external environmental factors had a relatively weaker influence on their behavioral safety decision-making. The majority of workers (about 91%) agree that management strategies have a positive impact on their behavioral safety decision-making, despite differences in the degree of acceptance. *II)* The findings revealed strong correlations among Questions 1–9 and Questions 11–15 and 17, indicating the rationality of the survey questionnaire design (see Appendix). While the majority of workers agree that management strategies have a positive impact on their behavioral safety decision-making, differences in the degree of acceptance exist.
- According to the responses of workers to Questions 20–30, we found that: *I)* Workers, in general, believe that physiological and psychological sensations, as well as management strategies, have a significant controlling effect on their unsafe behavior decisions (as seen in Questions 20–22). Only two workers in Questions 29–30 added other influencing factors, including the behavior of other workers and the overall safety atmosphere. This indicates that the model structure we

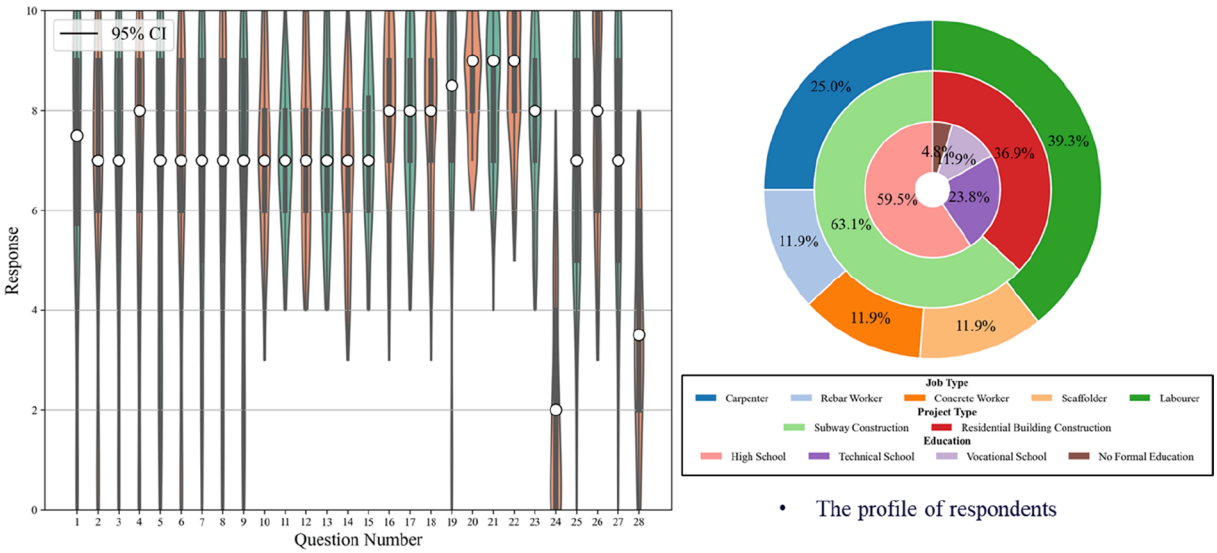


Fig. 6. Box Plot of Score Differences among Survey Respondents.

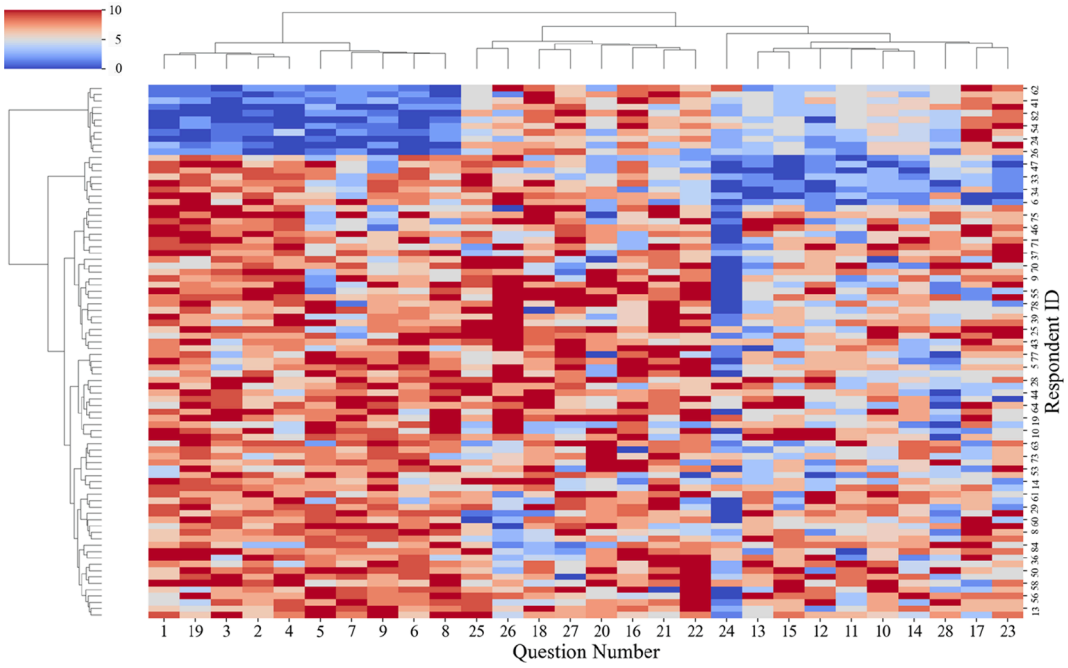


Fig. 7. Hierarchical clustering heatmap of scores among respondents to survey questionnaire.

proposed and the major influencing factors involved are fundamentally reasonable. *II)* Specifically, although all these factors have a significant impact on workers' unsafe behavior decisions, their impact mechanisms differ. For instance, 94% of the respondents think that physiological and psychological sensations have a more direct effect on inducing unsafe behavior (as discussed in Question 26). Meanwhile, 93% of the participants believe that management measures related to safety culture development have a cumulative effect on improving safety behavior levels over time (as discussed in Question 24). 86% of the respondents also think that reward and punishment measures have a relatively immediate effect on improving safety behavior performance (as discussed in Question 25). Moreover, 76% of the workers think that management measures related to safety culture development are more likely to be accepted by workers and may achieve more long-term management effects (as discussed in Questions 27–28). These findings suggest that the model proposed in this paper is reasonable, as demonstrated in Section 3.2. *III)* Furthermore, 90% of the respondents think that individual workers' unsafe behavior decisions exhibit biases, randomness, and uncertainty, even under the same environment and management strategy, as discussed in Question 23. These characteristics are consistent with the features reflected by the discrete choice model adopted in this paper to simulate workers' behavior choices in a specific situation.

Regarding the validation of our proposed model, the findings above provide strong evidence that the mathematical model presented in Section 3 methodology is appropriate for accurately reflecting the input factor structure and output behavior characteristics of workers' unsafe behavior decisions.

I) The approval and recognition of the structure of our model by most respondents, as revealed by our on-site worker questionnaire survey, suggests that workers' decision-making regarding unsafe behavior is influenced by multiple factors, including physiological and psychological aspects of the work environment, as well as management strategies. This finding supports the consistency of our proposed model with real-world situations.

II) Furthermore, the recognition by the majority of respondents in the survey of the regularities and characteristics of unsafe behavior formation in our model further validates its effectiveness. The survey questionnaire provided ample validation for various characteristics captured in our model, such as individual differences and randomness of workers' behavior, as well as the differences in timely effects caused by disruptive stimuli and management measures.

Overall, these findings suggest that our proposed mathematical model effectively captures the complex interplay between various factors influencing workers' decisions to engage in unsafe behavior, and thus provides valuable insights for developing effective interventions aimed at reducing unsafe behavior in the workplace.

5. Simulation experiment

5.1. Simulation process

This study used the hybrid simulation model to explore the mechanism by which unsafe worker behavior is formed as a result of environmental factors, management measures and psychological processes. The simulation experiment in this study attempts to probe into the following problems:

I) In the context of Dual Attitude theory, how does a worker deal with the conflicts between his unsafe impulses and professional ethics?

Simulation Step:

1.1) Develop a hybrid simulation model for a worker's unsafe behavior based on the theory in Section 3.2.1 and 3.2.2.

1.2) Test the effects of environmental perturbations on a worker's unsafe behavior rate.

1.3) Test the effects of management frequency on a worker's unsafe behavior rate.

II) In the context of Reinforcement Learning theory, how does a worker behave under a high pressure of safety management?

Simulation Step:

2.1) Develop a hybrid simulation model for a worker's unsafe behavior based on the theory in Section 3.2.3.

2.2) Test the effects of punishment strictness on a worker's unsafe behavior rate.

2.3) Test the effects of discount factor on a worker's unsafe behavior rate.

III) Comparing proactive management with reactive management, what are their features and how are they different?

Simulation Step:

3.1) Combine the simulation models of proactive management and reactive management.

3.2) Test the features of proactive management: leave the construction site in an uncontrolled condition for a period of time, take proactive actions at time t_1, and then remove the actions at time t_2.

3.3) Test the features of proactive management: leave the construction site in an uncontrolled condition for a period of time, take reactive actions at time t_1, and then remove the actions at time t_2.

3.4) Compare the safety performance in step 3.2) and 3.3).

5.2. Model initialization

The hybrid simulation model for worker's unsafe behavior consists of three parts, i.e., the control panel (from the Project Manager (PM)'s point of view), the proactive management panel (from a worker's point of view) and the reactive panel (from a worker's point of view).

I) The control panel. As shown in Fig. 8, the control panel simulates the PM's management role on the construction site, including policy release, on-site monitoring and the permissibility of penalty and possibility of discharge.

In the administrator module, various management policies and measures can be formulated freely. In the context of proactive management, how often pre-control and preventive measures are carried out for workers can be determined using the slider. The number at the bottom left indicates the minimal value of the slider, the number at the bottom right indicates the maximum value, and the number at the bottom center shows the current value selected on the slider. In the context of reactive management, the punishment strictness and the discount factor can be adjusted. Also, the frequency of occurrence of environmental perturbations, the main causes for unsafe-behavior impulse, can be customized by the software users. The same applies to the site environmental parameters, including the environment hazard level and the exposure probability, which are important parameters greatly affects the number of accidents given a certain number of unsafe behaviors.

In the statistics module, the safety performance of all workers is aggregated into several indicators, which represents the overall safety management state of the construction site, including the number of unsafe behaviors, the number of near miss incidents, the number of minor injuries, and the number of major injuries. All these indicators are critical in formulating reactive management measures for PM, e.g., penalty and possibility of discharge intensity.

In the punishment module, the intensity of the punishment for a worker's detected unsafe behavior or safety accident is determined based on the recent safety performance of the construction site. If the safety performance of the overall construction site is very poor in recent days, severe punitive measures will be taken to create a pressing stance against violations, with the aim of preventing workers from behaving in an unsafe manner. Conversely, minor punitive measures will be taken if the safety performance of the construction site is not so bad.

The specific modeling steps of the statistics module and the punishment module have been described in Section 3.2.3.

II) The proactive management panel. As shown in Fig. 9, the proactive management panel simulates the worker's psychological process from a dual attitude perspective. The panel consist of three parts, including Impulsive Precursors, Reflective Precursors, and Accident Generation.

In the context of reflective precursor module, the effects of safety management measures on the self-regulating character of workers are analyzed. As shown in Table 1, the key variables in this module include $y_{g,t}$ (the growth rate of y_t), $y_{d,t}$ (the decrease rate of y_t), and y_t (the value of safety awareness, safety proactivity, or emotional exhaustion at time t). By continuously updating these variables, we can obtain workers' reflective cognitions. The specific calculation steps are as follows: 1. Implementing various management measures leads to an increase in y_t, denoted as $y_{g,t}$; Over time, y_t exhibits a gradual decline, denoted as $y_{d,t}$. As a consequence, the formula for $y_{t+1} = y_t + y_{g,t} - y_{d,t}$; 2. The relationship between $y_{g,t}$ and the implementation of management measures has been described in Eq. (3), which is primarily based on learning curve theory [63]. 3. The declining pattern of $y_{d,t}$ over time has been stated in Eq. (4), which is primarily based on forgetting curve theory [64]. 4. By integrating individual workers' safety awareness, safety proactivity, and emotional exhaustions (considering that excessive control measures

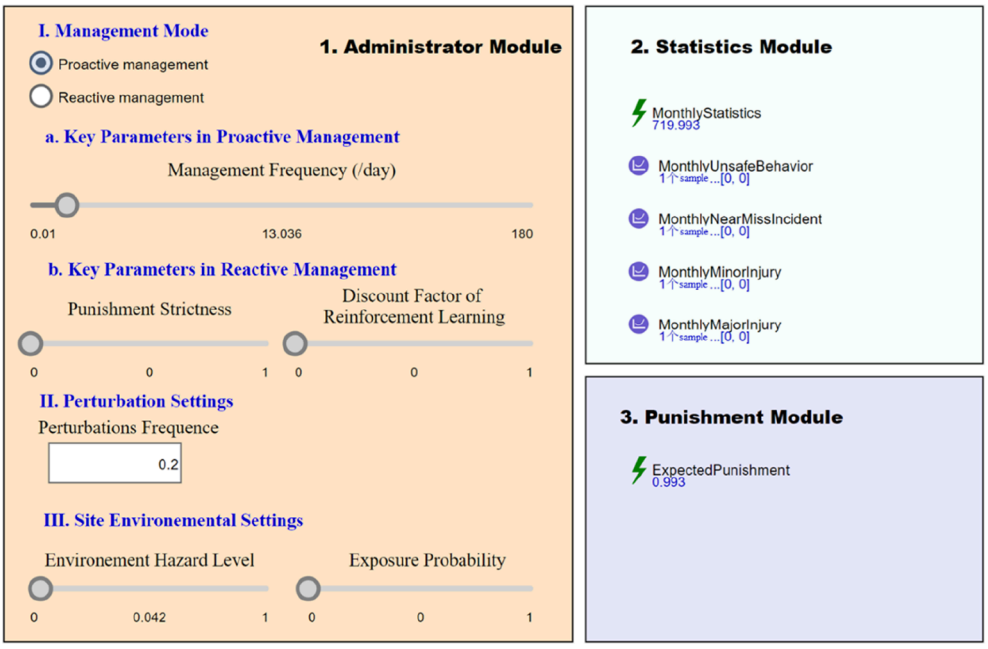


Fig. 8. Control panel in the software interface.

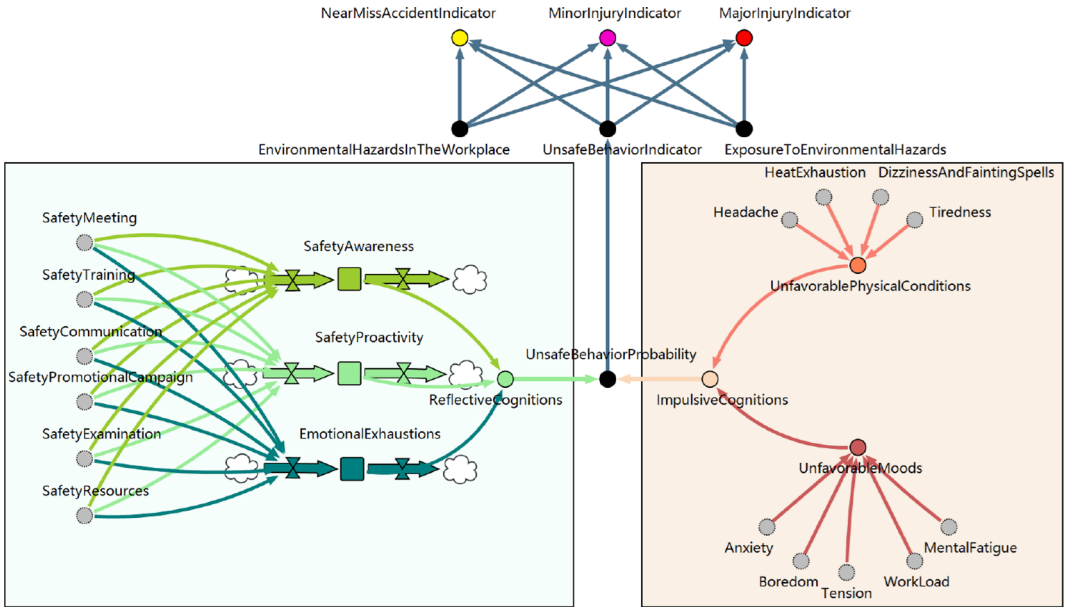


Fig. 9. Psychological process of unsafe worker behavior based on dual attitude theory.

may induce fatigue and numbness among workers, leading to adverse effects), the Reflective Cognitions of individual workers can be computed.

In the context of impulsive precursor module, an unfavorable external environment can disturb the physical and mental conditions of a worker. In an unfavorable physical and mental state, workers are prone to boredom and shallow labor, which leads to an impulsive thought of doing dangerous behaviors. As shown in Table 1, The key variable in this module is q_t (the numerical value of adverse physical conditions or negative emotions at time t), and the computation steps are as follows:1. Compute $q_t = \sum_{i=1}^{6} \beta_i \bullet v_i$ using a linear function. According to the dual attitude theory [62], the hot system (impulsive attitude) is triggered automatically by the external environment without much cognitive deliberation, whereas the cool system requires a more complex cognitive process to enact self-control. Thus, the formation of impulsive attitude U_r is straight-forward and does not use working memory, so the linear functions are used here to model the causal relationship between his impulsive cognitions and the influential factors; 2. The impulsive cognitions of individual workers can be computed by integrating their unfavorable physical condition and unfavorable mood.

The biggest difference between the impulsive precursor module and the reflective precursor module is: The impulsive cognitions forms and changes quickly in response to stimuli from the environment, whereas reflective cognitions involve a growing process that cannot be done in a single step. Thus, the impulsive precursor module is developed mainly on the basis of auxiliaries and their relational functions, and the reflective precursor module is developed mainly on the basis of flow,

level and rate (note that 'flow', 'level', 'rate', and 'auxiliary' are the terminologies in System Dynamics Modeling).

In the context of the Accident Generation module, the probabilities of unsafe behavior, near miss, minor injury and major injury are estimated. As shown in Table 1, whether these events occur or not will be determined by their probabilities using a random number. Suppose $o \in [0, 1]$ is a random number generated by computer, $k \in [0, 1]$ is the probability of a certain event (e.g., unsafe behavior, near miss, minor injury or major injury). If $o < k$, the corresponding event occurs. Otherwise, the corresponding event will not occur.

III) The reactive management panel. As shown in Fig. 10, the reactive management panel simulates the worker's psychological process from a reinforcement learning perspective. The panel consist of two parts, including the management environment and the worker agents.

In the context of worker agent module, every worker makes their decision based on the principle of minimal loss. As shown in Table 2, the module calculates the worker's safety performance ($SafetyPerformance_{i,t}$) based on major injuries and unsafe behavior indicators. The unsafe behavior indicator ($UnsafeBehaviorIndicator_{i,t}$) is determined by comparing the expected punishment ($Punishment_{i,t}$) with the worker's impulsive cognitions ($ImpulsiveCognitions_{i,t}$). Workers engage in unsafe behavior if the expected punishment is lower than their impulsive cognitions. When the expected punishment is harsh, workers compromise between loss and benefit and refrain from engaging in dangerous behavior.

In the context of the management environment module, the expected punishment is calculated based on Eq. (6), which represents the project manager's punishment policy aimed workers' unsafe behaviors and their consequential accidents. As shown in Table 2, this module calculates the expected punishment ratio ($PunishmentRatio_t$) based on the safety performances of all workers. The punishment ratio represents the level of punishment strictness imposed by the project manager. According to Reinforcement Learning [66], $PunishmentRatio_t$ represents the worker's expected consequences of their behavior based on the recent safety management situation, which is calculated as the cumulative discounted penalty resulting from their actions.

5.3. Results

Given the complete model described in the above sections, the simulation experiment can be started to verify the questions proposed in Section 4.1.

Here, we built a virtual construction site with 84 construction workers. The personal characteristics of these workers are collected using the survey questionnaire that has been detailed in Section 4, so as to imitate the actual situation of the diverse and uncertain personalities of construction workers. The sensitivity coefficients of workers to physical environment, psychological environment, and management measures, as shown in Fig. 8, correspond to questions 1–16 in the survey questionnaire in Appendix. It is noteworthy that the original evaluation scores provided by respondents will undergo standardization (with a minimum value of 0 and a maximum value of 1) before being entered into the model. For each simulation experience, we will first define the key parameters, then let the model run under the predefined parameters for a period of time. In this process, we explore and quantify the functions of these key parameters on the overall system by observing the changes in the model.

To provide a more vivid and illustrative representation of the workers' conditions, a visualization tool (Anylogic) is utilized to demonstrate the dynamics of the on-site situation during the simulation process. Workers are represented as nodes within the virtual site, with different colors representing their respective statuses. Fig. 11 exemplifies the simulated effects of the proposed model. This figure displays the changes in the safety behavior of all workers during a simulation run, with each worker represented by a node of a different color. For instance, a red node signifies a worker's involvement in a major injury accident, while a purple node represents a minor injury accident. A yellow node indicates a worker's involvement in a near-miss incident, a black node suggests unsafe behavior without resulting in an injury accident, and a consistently green node indicates the worker's adherence to safe practices throughout the day. It should be noted that due to space constraints, Fig. 11 only displays the visual scenes for specific days (0, 5, 10, 20, 50, 200) of a simulation run as an illustration. For comprehensive safety performance data of worker behavior on all other dates, refer to Figs. 12-15.

Three predefined experiments, namely Experiment A, B, and C (As shown in Table 3, 4, and 5), were designed to analyze the behavioral safety performance of workers under different management strategies.

Experiment A. Analysis of differences in workers' behavioral safety performance.

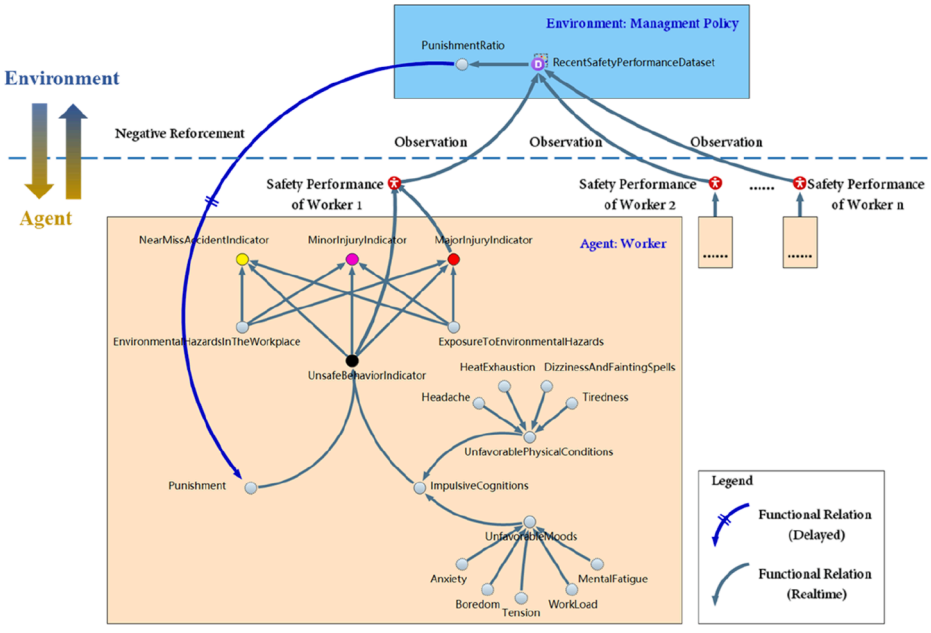


Fig. 10. Psychological process of unsafe worker behavior based on RL theory.

Table 2
Equations used in the reactive management panel.

Module	Model variables	Equations
Worker Agent module (Theory basic:[66])	$SafetyPerformance_{i,t}$ (Safety Performance of worker i at time t)	$SafetyPerformance_{i,t} = (1 - Punishmentstrictness) \bullet MajorInjuryIndcator_{i,t} + Punishmentstrictness \bullet UnsafeBehaviorIndicator_{i,t}$
	$UnsafeBehaviorIndicator_{i,t}$(Unsafe behavior indicator of worker i at time t)	$UnsafeBehaviorIndicator_{i,t} = \begin{cases} 1, Punishment_{i,t} < ImpulsiveCognitions_{i,t} \\ 0, Punishment_{i,t} \geq ImpulsiveCognitions_{i,t} \end{cases}$
	$Punishment_{i,t}$ (Expected punishment for worker i at time t if he takes the unsafe behavior)	$Punishment_{i,t} = PunishmentRatio_{t-1}$
Management Environment module (Theory basic:[66])	$PunishmentRatio_t$	$PunishmentRatio_t = \gamma \bullet PunishmentRatio_{t-1} + \sum_{i=1}^{n} SafetyPerformance_{i,t}/n$

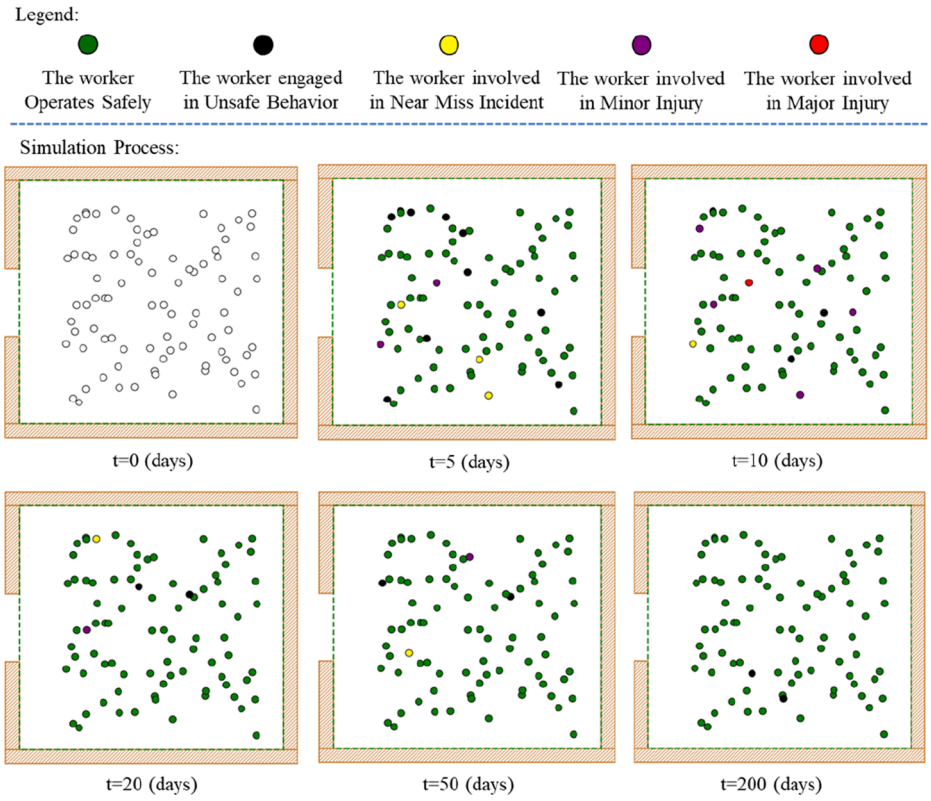


Fig. 11. Visualization and simulation of workers' unsafe behaviors on construction site.

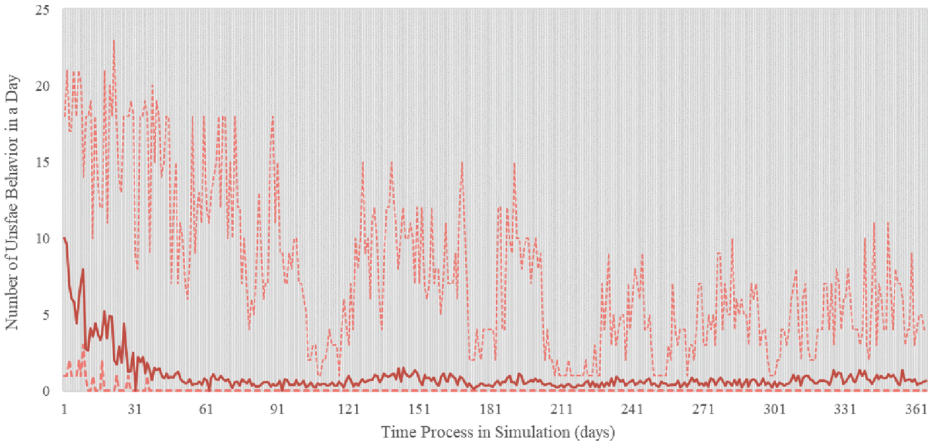


Fig. 12. Functions of perturbation frequency on behavioral safety performance.

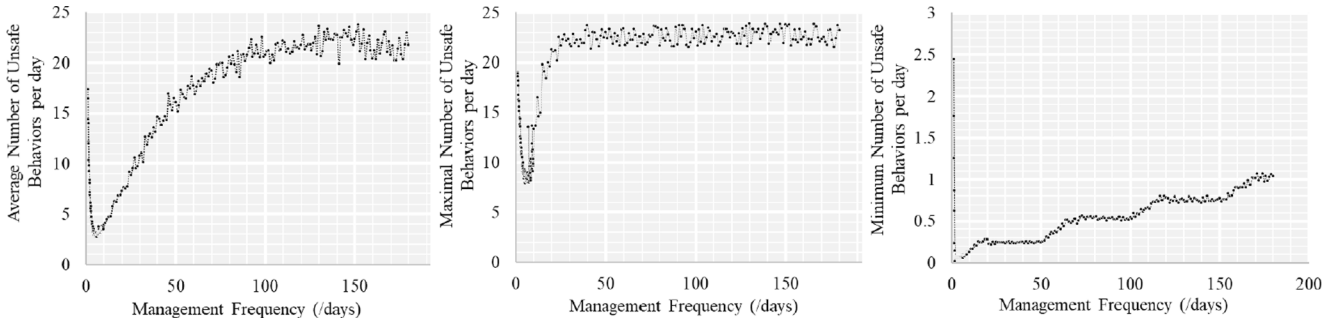


Fig. 13. Functions of intervention frequency on behavioral safety performance.

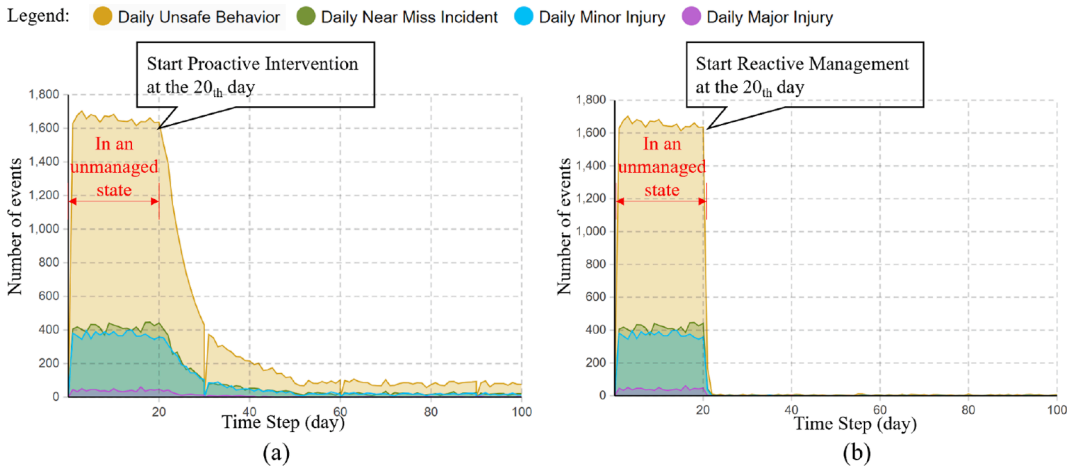


Fig. 14. Results of the first comparative experiment.

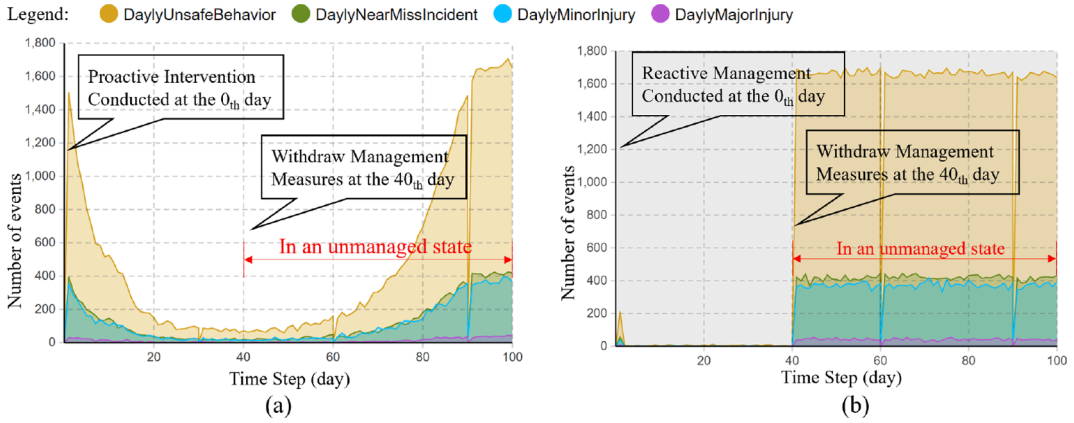


Fig. 15. Results of the second comparative experiment.

Under the above settings, the dynamic safety performance in a predefined simulation experiment can be output through observation. As shown in Fig. 12, three representative curves reveal the safety progress of the whole construction site. Since the personality of workers throughout the construction site is different, some workers may have strong self-discipline, and some workers are accustomed to unsafe behavior. As a result, even on the same worksite and in the same environment, different people will have different safety performances, which leads to these three curves. In Fig. 12, the top curve represents the number of unsafe behaviors of the worker with the highest total number of unsafe behaviors (among 84 workers), and shows how the number of unsafe behaviors changes under a given management strategy, while the bottom curve represents the safety performance of the worker with the strongest self-discipline. Different from the two extreme workers mentioned above, the average number of unsafe behaviors stand for the safety performance of the majority of workers, representing the general level.

Experiment B. Testing the functions of intervention frequency (/days).

The results of Fig. 13 show that reasonable safety management measures play an important role in improving workers' behavior safety capacity. Combined with the dual attitude theory and ordinary engineering experience, the results in Fig. 13 are reasonable. Proper strengthening of safety management promotes the development of a safe construction atmosphere and improves worker safety awareness and capacity. On the contrary, loose discipline brings slack and irregular

Table 3
Setting of Experiment A.

Management Mode	Proactive management
Management frequency	Intervention every 7 days
Perturbation frequency	Perturbation every 0.2 h
Environment hazard level	0
Exposure probability	0

behavior to workers.

Furthermore, the impact of the frequency of interventions on workers' safety performance is non-linear. Take the graph of 'Average Number of Unsafe Behaviors' in Fig. 13 as an example: when the intervention interval exceeds 50 days, the behavioral safety performance of the workers does not change much with the intervention interval. From once every 7 days to once every 50 days, workers' behavioral safety performance increases rapidly as the intervention interval is reduced. This shows that the construction of safety awareness and capacity is not dispersed and stacked, and it needs a systematic and ongoing learning process to have its effect. This pattern is more evident in the graph of 'Maximal Number of Unsafe Behaviors'.

Additionally, an abnormal trend in negative growth in safety performance was observed between once a day and once every 7 days. This trend reflects the need to keep response management measures at a reasonable level. When intervention measures reached a certain intensity, safety awareness and worker capacity were progressively built. After exceeding a certain threshold, excessive management measures can increase workers' workload, cause workers mental fatigue and low safety performance. This influence is consistent with typical work experience on a construction site.

Experiment C. Comparing proactive versus reactive management features.

In Experiment C, the influence mechanism between proactive intervention and reactive management is compared and summarized. Table 5 shows the parameter values in different management mode. Based on the experimental configuration of Table 5, comparative experiments are performed to distinguish the characteristics of these two modes.

As shown in Fig. 14, in the first 20 days, no management measures were taken. In the next 80 days, the proactive intervention and the reactive management strategy were adopted respectively. By comparing Fig. 14(a) with 14(b), it is found that reactive safety measures can have an immediate and intensive impact on workers' safety performance, while proactive interventions have a slower and relatively mild impact.

Table 4
Setting of Experiment B.

Management Mode	Proactive management
Management frequency	A dynamic variation from 'intervention every day' to 'intervention every 180 days'
Perturbation frequency	Perturbation every 0.2 h
Environment hazard level	0
Exposure probability	0

Table 5
Setting of Experiment C.

Proactive Intervention	**Reactive Management**
• Perturbation Time Interval = 0.2 h. Intervention Time Interval = 7 days. Environment Hazard Level = 0.5. Exposure Probability = 0.5. Number of Worker Agents = 84	• Perturbation Time Interval = 0.2 h. Severity of Punishment = 1. Discount Factor of Reinforcement Learning = 0.9 • Environment Hazard Level = 0.5. Exposure Probability = 0.5. Number of Worker Agents = 84

Expect the difference in efficiency, both of them can keep the frequency of workers' unsafe behaviors at a reasonably low level.

The results in Fig. 15 have shown that proactive interventions and reactive management measures can effectively maintain safety performance at an acceptable level. If we take away the corresponding measures, how is the safety performance of the construction site going to change? The results in Fig. 15 provides a preview of probable development process to this question. Proactive interventions and reactive management measures were adopted from the first day to the 40th day respectively in Fig. 15(a) and 15(b). When the management measures were withdrawn, they had completely different reactions. In the proactive control mode (as shown in Fig. 15 (a)), workers can still maintain the initial safety level for a long period of time. In comparison, in reactive management mode, once the reinforcement measures are removed, the workers' safety performance will deteriorate at an extremely fast pace. Clearly, proactive management provides a long-term guarantee of worker safety rather than reactive management.

6. Discussion

6.1. The rationale behind method selection

Based on the actual investigation and data statistics in Section 4, we discovered that unsafe behavior among workers results from a combination of physiological, psychological, and managerial factors. While workers exhibit consistent behavioral tendencies in similar situations, there is still a high degree of uncertainty and randomness, making it difficult to establish a clear, predictable, repeatable, and stable causal relationship between input (scenarios) and output (behaviors).

After comparing the results from Sections 4 and 5, we found that the empirical study findings align with the assumptions and compositions of our proposed model, making it an effective tool to explain real-world phenomena. For example, Fig. 12 simulations reveal significant variations in workers' safety performance, despite being under the same management strategy. While the number of unsafe behaviors decreases, the extent, speed, and fluctuation differ significantly. This individual variation in safety behavior is consistent with the results presented in Fig. 6, as observed in the survey questionnaire analysis. Furthermore, the simulations in Figs. 14 and 15 show that different management strategies have varying effects. Proactive management strategies, such as safety meetings and training, are progressively improving worker safety awareness and skills. This approach is slower, but it usually has a more sustainable impact, as workers are more receptive to it. In contrast, reactive management through strict rewards and penalties yields immediate results, but has a relatively short-term impact. These findings align with the tendencies observed in the answers to questions 16–19 and 24–27 in Fig. 6. Given the consistency between the simulation experiment findings and the survey questionnaire statistics, we can conclude that our model is not only capable of reflecting certain patterns and characteristics of dangerous behaviors in the real world, but also has the ability to simulate the evolutionary process of scenarios related to worker behavior.

Based on the results of the above experiments and literature review, we further discuss the rationale for using simulation methods to study unsafe behavior among workers. Currently, research on unsafe behavior among workers can be divided into two categories, i.e., data-driven causal analysis methods and ABM methods. Despite the limitations of simulation methods in reliability and predictive capability, it has benefits in interpretability, flexibility, and close integration with theories of behavioral psychology. The following reasons explain why this paper uses multi-agent simulation methods to investigate unsafe behaviors in workers:

)I. ABM is a mainstream research method for human behavior. Unsafe behavior among workers is also a kind of human behavior, and ABM is currently one of the mainstream methods for studying

human behavior, widely used in emergency evacuation behaviors [70], transportation behaviors [71], bike-sharing behaviors [72], consumers' purchase behaviors [73], etc. Compared with other fields, unsafe behavior among workers is also a type of 'human behavior' and should not be an exception.

)II. The special nature of human behavior. Human behavior is a combination of rational and irrational factors, with certain rules but also strong randomness and uncertainty (generally not having a clear, predictable, repeatable, and stable input–output causal relationship). Unlike other physical phenomena with stable 'input-output' mapping relationships, human behavior as an 'output result' also depends on the external 'input stimulus' it receives (according to Lewin's field dynamics theory), but this mapping relationship is often not entirely stable and can easily lead to differentiated behavior in the same situation due to various irrational or unexpected factors. The ABM method is a reliable way to simulate this uncertain and random behavior through repeated Monte Carlo experiments.

)III. Workers' opportunistic behavior can hinder the collection of authentic psychological data. When workers are aware that they are being observed or monitored, they may consciously pay attention to the safety of their behavior, even if they do not genuinely want to behave safely. Once they are no longer being observed or monitored, they may revert to unsafe behavior due to a sense of complacency or convenience. Therefore, workers' opportunistic behavior under observation is not conducive to collecting data on their true psychological reactions. This study had to resort to anonymous questionnaires to investigate workers' true psychological activities and used simulation methods to replicate this process.

6.2. Practical use

The engineering significance of this study is that it emphasizes the importance of implementing proactive safety management measures in order to improve workers' safety awareness and reduce accidents on construction sites. The proposed visualization model can be instrumental in developing effective safety management strategies by simulating different scenarios and their potential outcomes. Ultimately, this approach can result in safer working environments for construction workers and a decrease in the number of accidents, injuries, and fatalities on construction sites. Fig. 16 depicts the step-by-step procedure for the practical implementation of the proposed model, which are shown below.

Step 1 involves the preliminary formulation of management strategies for ensuring the safety of construction workers, such as choosing between proactive or reactive control strategies, identifying specific measures for proactive control, and determining the frequency of control measures.

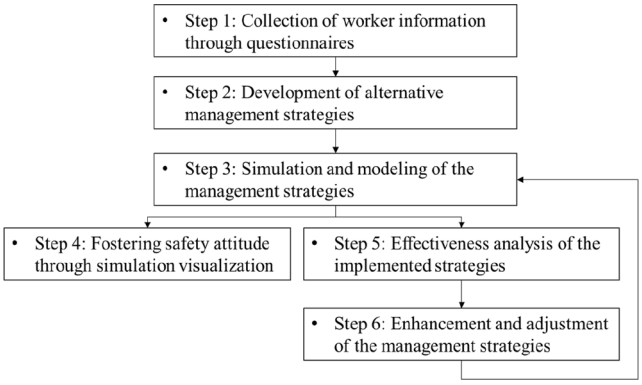


Fig. 16. Procedure for implementing the proposed model.

Step 2 involves collecting basic information about construction workers on-site and obtaining data on their decision-making tendencies and behavioral patterns under different environments and management strategies through the use of questionnaires.

Step 3 involves building a simulation model that can simulate changes in safety control on construction sites under given management strategies. The simulation includes tracking the number of unsafe behaviors as well as the frequency of near-miss incidents, minor accidents, and major accidents that result from these behaviors.

Based on the obtained simulation model, it can support the following two purposes which are also the research outcomes of this study:

I) Visualized Safety Management Training. Step 4 focuses on the need for proactive management measures to address the lack of workers' comprehension of safety behavior significance when they are compelled to act safely under monitoring pressure. This lack of understanding hinders the enhancement of overall safety awareness. Unfortunately, many construction projects prioritize safety only during accidents or special inspections, neglecting proactive management measures. To combat this, a visualization model capable of simulating the dynamics of behavior safety performance, and the resulting accidents under different control strategies is necessary. The outcomes of the simulation can be presented to decision-makers, allowing them to comprehend the importance of proactive safety management strategies and develop effective safety management approaches.

II) Enhance Existing Behavior Safety Management Strategies. Step 5 involves conducting a preliminary assessment of the management effectiveness of the pre-determined strategies on the current population of workers. In this step, the degree of external interference can be controlled, including physiological challenges arising from the work environment and unfavorable psychological states of individual workers. This allows for the observation and simulation of worker behavior safety performance under various possible working conditions. Step 6 allows management decision-makers to adjust and enhance existing management strategies based on the simulated safety performance results obtained under the current management strategy. Importantly, the dynamic fluctuation characteristics (which have mostly been verified in the analysis of the survey questionnaire results in Chapter 4) can illustrate qualitative trends and regularities to some extent, providing reference value for the real physical world.

6.3. Contribution

This paper presents an agent-based composite simulation model to construct and manage the simulation of unsafe behavior among construction workers. The model simulates how unsafe behavior is generated and managed at both the micro level of individual worker behavior decision-making and the macro level of management strategy development, as well as how management strategies affect worker behavior.

The contributions of this study lie in the following aspects.

I) Theoretical supports: It is supported by several behavioral psychology theories, such as the dual attitude theory, reinforcement learning theory, and discrete choice theory. These classic theories make the proposed method highly interpretable and non-black-box. Human behavior is not always rational, as evidenced by drug addicts who are aware of the harm caused by drug abuse but still engage in it, which is no exception for workers' unsafe behaviors. But existing research has largely ignored the relationship between the conflicts of primal impulses and rational inhibition, which has been presented in classic theory of behavioral psychology [74]. In addition, the use of reward and punishment measures as an efficient means of modifying human behavior has seldom been modeled or explored in previous studies of workers' unsafe behavior. However, it is clear that there are significant differences between the Skinner's behavioral reinforcement theory [75] and the dual attitude theory [36] in determining human behavior, and they should be separately considered and modeled. This study establishes simulation models for both proactive and reactive safety management

strategies and compares their safety performance under different management scenarios, providing valuable information for the development of effective safety management strategies.

II) Combination of simulation and real-world data: To ensure authenticity and reliability of psychological data collected from construction workers, it may be beneficial to use anonymous questionnaires in order to avoid potential bias resulting from workers' opportunistic tendencies. Based on this, simulation methods can serve as an important interface for accessing interdisciplinary knowledge and providing a non-invasive research environment. On one hand, incorporating real data obtained from surveys into the model allows for the assignment of personality features from actual workers to multiple agents without disrupting their normal work. This process facilitates the construction of artificial intelligent agents. On the other hand, behavioral psychology theories can be mathematically modeled to replicate workers' psychological activities and the process by which unsafe behavior arises. The benefit of combining simulation and real-world data is that it simultaneously takes into account the authenticity and reliability of the data, as well as the interpretability and exportability of the model. It crosses different levels of modeling, including both the inner activities of individual workers and the management strategies of worker groups. This allows researchers to explore different hypotheses and scenarios and observe their impacts intuitively. Because simulation methods allow researchers to experiment with multiple experiments and conditions, they can draw more accurate and reliable conclusions.

III) Valuable findings in line with real-world situations: Valuable and realistic findings can be obtained through simulation methods to explain the underlying mechanisms of unsafe behaviors exhibited by construction workers in real-world settings. At the micro-level, our model reflects the randomness and uncertainty of individual behavior decisions through discrete behavioral choices, which is consistent with the results obtained from real-world questionnaires. At the macro-level, workers' unsafe behaviors exhibit characteristics of a complex system that is self-adaptive and self-organizing, which is consistent with the dynamic adjustment features of actual engineering project management methods. This self-adaptive and self-organizing mechanism displays time-sensitive differences between proactive and reactive strategies regarding their impact on construction workers' safety performance during intervention and withdrawal. The research findings emphasize the critical role of proactive safety management in construction sites, providing decision-makers with valuable insights into effective safety management plans.

7. Conclusion

Ensuring workers' behavioral safety is a persistent challenge for on-site managers. Managers utilize a range of strategies, such as proactive interventions and reactive management. Proactive interventions, grounded in the dual attitude theory, gradually foster workers' awareness of safety. Once established, this awareness facilitates the maintenance of behavioral safety regardless of external influences. Driven by reinforcement theory, reactive management aims to attain immediate rewards. In urgent situations, such as accident-prone construction sites, implementing stringent measures is necessary to prevent a decline in safety performance. However, the removal of these measures may cause workers to become lax, resulting in an increase in unsafe behaviors and accidents.

This paper described a hybrid simulation method to explore the formation mechanisms for unsafe worker behaviors as well as the effects of the management strategy. Managing and controlling unsafe worker behaviors is a dynamic, closed-loop and feedback process, which includes both individual subject activities at the micro level and performance emergence at the macro level. Benefitting from the hybrid simulation method, the process of behavioral reflection and decision-making can be modelled and analyzed in a quantified manner. Results from the simulation model provide valuable information for policy analysis and decision-making for project managers. The information provided by the model help them anticipate and understand how workers respond in thought and action as part of their management policies.

Due to various constraints, such as respondent willingness and potential pandemic restrictions, the data collected from the survey questionnaire phase of our study was limited. However, we implemented measures to ensure data reliability and believe our findings are still representative. For future research, it is recommended to address the limitation by increasing the sample size to strengthen the generalizability of the findings. Additionally, incorporating advanced equipment such as EEG (electroencephalography) and eye-tracking devices can be considered to collect psychophysiological information related to workers' unsafe behaviors. This would facilitate a deeper exploration and confirmation of the factors influencing unsafe behaviors, the formation mechanisms, and the interactive relationships among workers and between workers and management. Moreover, it is important to build upon the current focus on individual internal psychological processes and the response relationship between individuals and policies by further considering the interaction among agents in the future. These advancements would provide a more comprehensive understanding of the topic and contribute to the field of behavioral safety management.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

A questionnaire investigating the driving forces and management strategy influences on workers' safety behavior

a) **Driving forces from Working Environment**

1. When you feel headache or discomfort, would you be more prone to making mistakes that could lead to unsafe behavior (such as mishandling equipment, not following safety procedures, etc.)?2. When working in extreme heat and sweating heavily, would you remove your hard hat?3. When you feel overworked and experience dizziness, would you be more

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likely to make mistakes that could lead to unsafe behavior (such as misjudging distances, not securing equipment properly, etc.)?4. When you feel physically tired, would you choose to take shortcuts and ignore safety rules (such as not using proper lifting techniques, not wearing appropriate PPE, etc.)?5. When you feel anxious, would you choose to take shortcuts and ignore safety rules in order to make progress (such as rushing through tasks without proper safety checks, not following lockout/tagout procedures, etc.)?6. When you feel bored and annoyed by tasks, would you choose to take shortcuts and ignore safety rules for progress (such as not properly securing equipment, not following proper procedures for hazardous materials, etc.)?7. When feeling unfocused, would you be more likely to make mistakes that could lead to unsafe behavior (such as not paying attention to your surroundings, not following proper procedures for operating machinery, etc.)?8. When given a tight deadline to complete a task, would you choose to take shortcuts and ignore safety rules in order to make progress (such as not wearing proper PPE, not properly securing equipment, etc.)?9. When feeling mentally distracted or nervous, would you be more likely to make mistakes that could lead to unsafe behavior (such as ignoring safety warning signs, not paying attention to the surrounding environment, etc.)?

Influences of Management Strategy

10. Have you noticed an improvement in your safety behavior after receiving detailed and sufficient technical briefing and safety meetings?11. Do you believe that safety training has a significant impact on improving your safety behavior?12. Do you believe that workplace safety reminder signs positively influence your safety behavior?13. Do you believe that safety communication with project managers has a significant impact on improving your safety behavior?14. Do you believe that safety inspections and reviews have a significant impact on improving your safety behavior?15. Do you believe that having sufficient safety resources, such as PPE, positively impacts your safety behavior?16. When you realize that you are being observed or monitored, do you consciously give more attention to the safety and normalization of your behavior?17. When you have been criticized, penalized, or had your bonus or salary deducted for unsafe behavior, would you become more attentive to standardizing your behavior?18. When you hear that there will soon be a special inspection targeting unsafe behavior, would you consciously pay more attention to safety and normalizing your behavior?19. When a special inspection related to unsafe behavior is completed, do you immediately relax and lose interest in the safety of your behavior?

Overall Regularity of The Model

20. To what extent do you believe the external work environment contributes to your commitment to unsafe behavior?21. To what extent do you believe the internal psychological states contributes to your commitment to unsafe behavior?22. To what extent do you believe the safety management strategy contributes to your commitment to unsafe behavior?23. Based on your previous work experiences, to what extent do you believe that the occurrence of unsafe behavior may still be contingent and random, despite the above influences?24. Do you believe that safety culture enhancement initiatives, such as safety training and safety meetings, have an immediate effect on improving your safety behavior?25. Do you believe that strict reward and punishment measures have an immediate effect on improving your safety behavior?26. If the factors mentioned in Questions 1–9 had an impact on leading to your unsafe behavior, to what extent do you believe that this driving effect is immediate and direct, rather than delayed?27. Would you be willing to accept safety culture measures like safety training and safety promotion campaign to improve your safety behavior?
28. Would you be willing to accept strict reward and punishment measures to improve your safety behavior?29. Apart from the above questions, what other factors do you think could significantly improve your decision-making regarding safety behavior?30. Apart from the above questions, what other factors do you think could significantly deteriorate your decision-making regarding safety behavior?

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